Elliptic Curve Codes: Recent Constructions and Isogeny-Induced Transformations

Jamie Low Jiaqi & Oliver Ang Wei Jie
School of Physical and Mathematical Sciences, Nanyang Technological University, Singapore

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Abstract

Elliptic curves are the only algebraic curves isomorphic to their own Jacobians, so their points admit a canonical group structure. When constructing algebraic geometry codes using elliptic curves, one can exploit the group law to obtain better parameters and prove certain bounds. This report introduces the machinery behind elliptic curve codes and surveys recent constructions of elliptic curve codes in both the classical and quantum settings. As a novel direction, we also explore how isogenies interact with elliptic codes and their parameters.

1 Introduction

Algebraic geometry (AG) codes are a family of linear error-correcting codes introduced by Valeriy Goppa in the early 1980s [1]. Broadly speaking, they are defined by evaluating rational functions on an algebraic curve $\mathcal{C}/\mathbb{F}_q$ at some set of evaluation points. They generalise the widely used Generalised Reed–Solomon (GRS) codes [2], which are obtained by letting $\mathcal{C}$ be the projective line $\mathbb{P}^1(\mathbb{F}_q)$, which has genus 0.

The code parameters are closely related to the geometry of the chosen curve, particularly its genus and the number of rational points. AG codes therefore provide a natural connection between algebraic geometry and coding theory. In 1982, Tsfasman, Vlăduţ and Zink took this idea further, using curves with many rational points to construct the first code families whose asymptotic parameters surpass the Gilbert–Varshamov bound [3], demonstrating the ability for algebraic geometry to produce asymptotically good codes and sparking further research of AG codes constructed from a diverse range of algebraic curves.

The simplest AG codes inequivalent to GRS codes arise from elliptic curves, which are smooth projective curves of genus 1. They are the only algebraic curves whose points admit a canonical abelian group structure [4, §X.3].[1] AG codes defined using elliptic curves are called elliptic curve codes, or simply elliptic codes. Using the group structure, algebraic constructions unique to genus 1 have been discovered, yielding codes with better parameters (e.g. maximum distance separable (MDS) codes) and desirable properties like iso-duality, with applications in quantum error correction. The group structure also supplies arguments for proving bounds on code parameters.

Elliptic curves also carry isogenies, which are curve morphisms preserving the group structure. Isogenies are heavily used in post-quantum cryptography [5–7], but their applications to coding theory remain largely unexplored. We embark on a computational project, using SageMath to explore the different ways in which isogenies can transform an elliptic curve's function field, and hence the parameters of an elliptic code.

The goal of our report is to explain how the remarkable properties of elliptic curves yield unique and excellent constructions of elliptic codes that stand out from general AG codes. In Sections 2 and 3, we introduce elliptic codes as a concrete and accessible instance of AG codes by developing the necessary background underlying the arithmetic of elliptic

[1] Provided they have an $\mathbb{F}_q$-rational point, elliptic curves are the only algebraic curves isomorphic to their own Jacobians and hence are abelian varieties.

curves and elliptic codes. The bulk of our report (Sections 4 to 6) constitutes a survey on recent constructions, bounds and applications. We conclude with a documentation of our exploration on the interaction between isogenies and elliptic codes in Section 7.

Due to space constraints, proofs of cited results are either postponed to the appendix or omitted, and appear in the main text only when they illustrate how the group law is used in a code construction.

2 Preliminaries

We summarise key definitions and results pertaining to elliptic curves and elliptic codes. For a more detailed description with proofs of results, we refer the reader to the following comprehensive standard references, which our exposition is based on:

- Silverman's book [4] for basic arithmetic geometry, with a focus on elliptic curves;
- Stichtenoth's book [8] for algebraic function fields and algebraic geometry (AG) codes.

Henceforth, let $q = p^m$ denote a prime power.

An **elliptic curve** $\mathcal{E}$ over $\mathbb{F}_q$ is a smooth projective curve of genus 1. Equivalently, $\mathcal{E}$ is the set of points $(x, y) \in \overline{\mathbb{F}}_q^2$ satisfying an equation of the form

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in \mathbb{F}_q \quad (1)$$

together with a point at infinity ∞, provided that no point is singular. A point $P \in \mathcal{E}$ is **$\mathbb{F}_q$-rational** if $P \in \mathbb{F}_q^2 \cup \{\infty\}$. The set of $\mathbb{F}_q$-rational points is denoted by $\mathcal{E}(\mathbb{F}_q)$, and the set $\mathcal{E}(\mathbb{F}_q)\setminus\{\infty\}$ of affine $\mathbb{F}_q$-rational points by $\mathcal{E}(\mathbb{F}_q)^*$. Its cardinality is constrained by the Hasse–Weil bound [4, §V.1]

$$|\#\mathcal{E}(\mathbb{F}_q) - (q+1)| \le \lfloor 2\sqrt{q} \rfloor.$$

The points of $\mathcal{E}$ form an abelian group $(\mathcal{E}, \oplus)$ with identity ∞, where the group addition $\oplus$ is governed by the equations in [4, §III.2] and visualised in Figure 1. As is standard in the literature, $[m]P$ denotes the m-fold sum $P \oplus \cdots \oplus P$ for $m \in \mathbb{N}$, and $[-m]P := \ominus[m]P$.

A **divisor** of $\mathcal{E}$ is an element D of the free abelian group $\operatorname{Div}\mathcal{E} = \mathbb{Z}[\mathcal{E}]$ over the points of $\mathcal{E}$. In other words, it is a formal sum

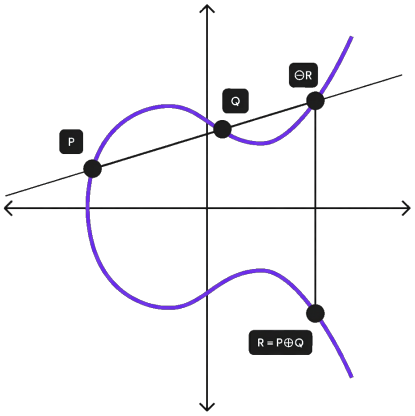


Figure 1. The group law of an elliptic curve $\mathcal{E}$ over $\mathbb{R}$ [9]

$$D = \sum_{P\in\mathcal{E}} n_P \cdot (P), \quad n_P \in \mathbb{Z}$$

where all but finitely many n_P are zero. Its **support** and **degree** are, respectively,

$$\operatorname{supp} D = \{P \in \mathcal{E} : n_P \neq 0\} \quad \text{and} \quad \deg D = \sum_{P\in\mathcal{E}} n_P.$$

Its **sum** is the evaluation of the formal sum using the elliptic group law, and is denoted by

$$\Sigma(D) = \bigoplus_{P\in\mathcal{E}} [n_P]P.$$

A partial ordering on $\operatorname{Div}\mathcal{E}$ is defined by

$$\sum_{P\in\mathcal{E}} n_P(P) \le \sum_{P\in\mathcal{E}} n'_P(P) \iff \forall P \in \mathcal{E} : \quad n_P \le n'_P.$$

The **function field** $\mathbb{F}_q(\mathcal{E})$ of $\mathcal{E}$ is the field $\mathbb{F}_q(x, y)$, where x is transcendental over $\mathbb{F}_q$ and y is algebraic over $\mathbb{F}_q(x)$ subject to Equation (1). For $f \in \mathbb{F}_q(\mathcal{E})$, set

$$\operatorname{div} f := \sum_{P\in\mathcal{E}} \operatorname{ord}_P(f) \cdot (P), \quad \text{where}$$

- $\operatorname{ord}_P(f) = n \in \mathbb{Z}^+$ if f has a zero of order n at P;
- $\operatorname{ord}_P(f) = -n \in \mathbb{Z}^-$ if f has a pole of order n at P;
- $\operatorname{ord}_P(f) = 0$ otherwise.

Such divisors are called **principal** and form a subgroup $\operatorname{PDiv}\mathcal{E}$ of the degree-0 divisors $\operatorname{Div}^0\mathcal{E}$. Two divisors (of any degree) are called **equivalent** if they differ by a principal divisor. The **Jacobian** of $\mathcal{E}$ is the quotient $\operatorname{Jac}\mathcal{E} := \operatorname{Div}^0\mathcal{E}/\operatorname{PDiv}\mathcal{E}$, which is famously isomorphic to $\mathcal{E}$ (cf. Section 4).[2]

Given a divisor $G \in \operatorname{Div}\mathcal{E}$, we construct the **Riemann–Roch space**

$$\mathcal{L}(G) := \{f \in \mathbb{F}_q(\mathcal{E})^* : \operatorname{div} f \ge -G\} \cup \{0\}.$$

[2] More precisely, this quotient is called the Picard group of degree 0, but this is isomorphic to the Jacobian variety.

This is a finite-dimensional vector subspace of $\mathbb{F}_q(\mathcal{E})$, and we denote its dimension by $\ell(G)$. For elliptic curves, the Riemann–Roch Theorem [8, §1.5] gives us $\ell(G) \geq \deg G$, with equality when $\deg G \geq 1$.

The **space of differential forms** $\Omega_\mathcal{E}$ on $\mathcal{E}$ is the $\mathbb{F}_q$-vector space generated by symbols of the form $\mathrm{d}f$ for $f \in \mathbb{F}_q(\mathcal{E})$, subject to the relations

$$\mathrm{d}(f+g) = \mathrm{d}f + \mathrm{d}g, \quad \mathrm{d}(fg) = f\,\mathrm{d}g + g\,\mathrm{d}f, \quad \mathrm{d}a = 0$$

for all $f, g \in \mathbb{F}_q(\mathcal{E})$ and $a \in \mathbb{F}_q$. Around a point $P \in \mathcal{E}$, every $\omega \in \Omega_\mathcal{E}^*$ can be written uniquely as

$$\omega = \sum_{n=m}^{\infty} a_n t^n \mathrm{d}t \quad \text{for some} \begin{cases} t \in \mathbb{F}_q(\mathcal{E}),\ \mathrm{ord}_P(t) = 1, \\ a_n \in \mathbb{F}_q,\ a_m \neq 0,\ m \in \mathbb{Z}. \end{cases}$$

For $\omega \in \Omega_\mathcal{E}^*$, set $\mathrm{ord}_P(\omega) := m$, $\mathrm{Res}_P(\omega) := a_{-1}$ (or 0 if $m \geq 0$),

$$\mathrm{div}\,\omega := \sum_{P \in \mathcal{E}} \mathrm{ord}_P(\omega) \cdot (P), \quad \text{and}$$
$$\Omega(G) := \{\omega \in \Omega_\mathcal{E}^* : \mathrm{div}\,\omega \geq G\} \cup \{0\}.$$

In fact, $\{\mathrm{div}\,\omega : \omega \in \Omega_\mathcal{E}^*\} = \mathrm{PDiv}\,\mathcal{E}$ [8, Cor. 1.4.12].

3 Elliptic codes

This section introduces general elliptic codes, their parameters, and their encoding and decoding mechanisms. In later sections, we consider specialised codes with better properties.

Let $n \in \mathbb{Z}^+$, $\mathcal{E}$ be an elliptic curve and $D, G \in \mathrm{Div}\,\mathcal{E}$ be two divisors such that $1 \leq \deg G \leq n-1$, $\mathrm{supp}\,D \cap \mathrm{supp}\,G = \varnothing$, and $D = (P_1) + \cdots + (P_n)$ is a sum of n distinct $\mathbb{F}_q$-rational points. We adopt the notation

$$f(D) := (f(P_1), \ldots, f(P_n)) \in \mathbb{F}_q^n \quad \text{for} \quad f \in \mathcal{L}(G).$$

Construct an ordered basis $(\varphi_1, \ldots, \varphi_k)$ of $\mathcal{L}(G)$, where $k = \ell(G) = \deg G$ by Riemann–Roch. Then, there exists an $[n, k, d]_q$ code

$$C_\mathcal{L}(D, G) := \{f(D) : f \in \mathcal{L}(G)\} = \langle \varphi_1(D), \ldots, \varphi_k(D) \rangle,$$

which encodes a message $\mathbf{m} \in \mathbb{F}_q^k$ as $\sum_{i=1}^k m_i \varphi_i(D)$.

The quantity $d^* := \deg(D - G) = n - k$ is called the **designed distance** of $C_\mathcal{L}(D, G)$. The code has minimum distance $d \geq d^*$ [8, §2], so the Singleton defect is at most 1. We will see in Section 5 ways to construct MDS elliptic codes.

The Euclidean dual code $C_\mathcal{L}(D, G)^{\perp_\mathrm{E}}$ of $C_\mathcal{L}(D, G)$ is

$$C_\Omega(D, G) = \left\{(\mathrm{Res}_{P_1}(\omega), \ldots, \mathrm{Res}_{P_n}(\omega)) : \omega \in \Omega(G - D)\right\},$$

as proven in [10, Prop. 8]. Choose $t \in \mathbb{F}_q(\mathcal{E})$ such that $\mathrm{ord}_{P_i}(t) = 1$ for $1 \leq i \leq n$. Then, the differential $\eta := \mathrm{d}t/t$ has the property that every $\omega \in \Omega(G - D)$ can be written as $f\eta$ for some $f \in \mathcal{L}(H)$, where $H = D - G + \mathrm{div}\,\eta$ [8, §8.1]. As such, we have the beautiful result that $C_\Omega(D, G) = C_\mathcal{L}(D, H)$. Consequently, $C_\mathcal{L}(D, G)$ is either MDS or near-MDS.

A common choice of G is $G = k(Q)$ for some $Q \in \mathcal{E}$. Such codes are called **one-point codes** and have the advantage that a basis of $\mathcal{L}(G)$ is easy to construct.

The remainder of this section is devoted to decoding. We illustrate one relatively simple (but suboptimal) decoding algorithm [11, §6.2][3] for the $[n, n-k, \geq k]$ code $C := C_\Omega(D, k(Q))$, the dual of a one-point code.

Denote by $R = \bigcup_{m=1}^\infty \mathcal{L}(m(Q))$ the $\mathbb{F}_q$-vector space of functions having no poles away from Q. Let $(\varphi_j)_{j \in \mathbb{Z}^+}$ be a basis for R, ordered such that $\varphi_1 = 1$ and $\mathrm{ord}_Q(\varphi_j) = -j$ for $j \geq 2$, so that $\mathcal{L}(m(Q)) = \langle \varphi_1, \ldots, \varphi_m \rangle$. For instance, if $Q = \infty$, we may take the basis $(1, x, y, x^2, xy, x^3, x^2y, \ldots)$.

Let $\mathbf{r} = \mathbf{c} + \mathbf{e} \in \mathbb{F}_q^n$ be a received word, where $\mathbf{c} \in C$ and $\mathbf{e} \in \mathbb{F}_q^n$ are an unknown codeword and error respectively. Define the **syndrome functional**

$$S : \mathcal{L}(G) \to \mathbb{F}_q, \quad f \mapsto \mathbf{r} \cdot f(D).$$

Let $f, g \in \mathcal{L}(s(Q)) \subset \mathcal{L}(G)$, where $s = \lfloor k/2 \rfloor$, so that $fg \in \mathcal{L}(G)$. Since $(fg)(D) \in C_\mathcal{L}(D, G) = C^\perp$, we have $\mathbf{c} \cdot (fg)(D) = 0$, and so

$$S(fg) = (\mathbf{c} + \mathbf{e}) \cdot (fg)(D) = \sum_{i \in \mathrm{supp}\,\mathbf{e}} e_i f(P_i) g(P_i).$$

Hence, the space

$$K = \left\{f \in \mathcal{L}(s(Q)) : S(fg) = 0\ \forall\, g \in \mathcal{L}(s(Q))\right\}$$

contains the functions f vanishing at $\mathrm{supp}\,\mathbf{e}$, and thus, reveals the error positions. Sampling $f \in K^*$, the coordinate set I where f vanishes contains $\mathrm{supp}\,\mathbf{e}$. As S is linear, it suffices to consider $S(\varphi_i \varphi_j)$ for $1 \leq i, j \leq s$.

Finally, we use [10, Prop. 6.1] to find the error values. Assuming $\#I < d(C)$, $\mathbf{e}$ is the unique solution $\mathbf{x}$ of

[3] This algorithm applies to higher-genus curves and does not use the group law. There are better algorithms [10, 12], but we select this one for pedagogical reasons.

the system

$$\mathbf{x} \star f_i(D) = S(f_i) \quad \forall\, 1 \le i \le k \quad \text{and} \quad x_i = 0 \quad \forall\, i \notin I.$$

[10, Prop. 6.5] shows that this decoding algorithm can correct $\lfloor k/2 \rfloor - 1$ errors (which is suboptimal if C is MDS) with complexity $\mathcal{O}(n^3)$.

Algorithm 2 (Basic Decoding [11, §6.2])

Input: A received word $\mathbf{r} \in \mathbb{F}_q^n$
Output: The decoded codeword $\mathbf{c} \in C_\Omega(D, G)$
Steps:

1. $\mathbf{S} \leftarrow [S(\varphi_i \varphi_j)]_{i,j=1}^{s} \in \mathbb{F}_q^{s \times s}$
2. $K \leftarrow \ker \mathbf{S}$
3. $f \leftarrow \sum_{i=1}^{s} \lambda_i \varphi_i$ for some $\boldsymbol{\lambda} \in K^*$
4. $I \leftarrow \{i : f(P_i) = 0\}$ // Contains $\operatorname{supp} \mathbf{e}$
5. $\mathbf{H} \leftarrow [\varphi_1(D), \ldots, \varphi_k(D)] \in \mathbb{F}_q^{n \times k}$
6. Let $\mathbf{e}$ be the unique (row) vector satisfying $\mathbf{eH} = \mathbf{rH}$ and $e_i = 0$ for all $i \notin I$
7. Return $\mathbf{r} - \mathbf{e}$

4 Exploiting the group law

Every smooth projective curve $\mathcal{C}$ of genus g carries an abelian variety of dimension g, its Jacobian $\operatorname{Jac}\mathcal{C}$, whose points are the divisor classes of degree zero on $\mathcal{C}$ (cf. Section 2). What distinguishes genus 1 is that the group structure can be transported from the Jacobian to the curve. Specifically, for an elliptic curve, the **Abel–Jacobi map**

$$\mathcal{E} \longrightarrow \operatorname{Jac}\mathcal{E}, \qquad P \longmapsto [(P) - (\infty)]$$

is a bijection whose inverse is the sum map Σ of Section 2, and the group law it induces on $\mathcal{E}$ agrees with the geometric one in [4, §III.3]. This is the isomorphism with the Jacobian noted in the abstract.

The consequence for coding theory is immediate. The code $C_{\mathcal{L}}(D, G)$ is built by evaluating functions at $P_1, \ldots, P_n$, and when $g = 1$ these points are themselves elements of a group. Conditions on D and G, which govern the code parameters, may therefore be rewritten as conditions on sums of these evaluation points using tools such as the Abel–Jacobi Theorem. For $g \ge 2$ the group lives on a higher-dimensional variety, and no such translation is available.

Lemma 3 (Abel–Jacobi [4, Cor. III.3.5])

$A \in \operatorname{Div}^0 \mathcal{E}$ is principal if and only if $\Sigma(A) = \infty$.

Henceforth we assume, as is standard, that $\operatorname{supp} G \subset \mathcal{E}(\mathbb{F}_q)$, so that $\Sigma(G) \in \mathcal{E}(\mathbb{F}_q)$. Section 3 established that an elliptic code is either MDS or near-MDS. The following criterion decides which.

Theorem 4 (Subset-sum criterion [13, 14])

Let $C_{\mathcal{L}}(D, G)$ be as in Section 3, with $D = (P_1) + \cdots + (P_n)$ and $k = \deg G$. Then $d = n - k$ if and only if there are k distinct points $P_{i_1}, \ldots, P_{i_k} \in \operatorname{supp} D$ with

$$P_{i_1} \oplus \cdots \oplus P_{i_k} = \Sigma(G). \tag{2}$$

Otherwise, $d = n - k + 1$, and the code is MDS.

Proof. See page 12 in the appendix. □

The criterion is more than a test, since its proof is constructive and hands us the minimum-weight codewords explicitly. Suppose $S = \{P_{i_1}, \ldots, P_{i_k}\} \subseteq \operatorname{supp} D$ satisfies Equation (2). The divisor $\sum_{P \in S}(P) - G$ then has degree 0 and sum ∞, so Lemma 3 produces a function f with $\operatorname{div} f = \sum_{P \in S}(P) - G$. This f lies in $\mathcal{L}(G)$ and vanishes on S and nowhere else in $\operatorname{supp} D$, so $f(D)$ is a codeword of weight exactly $n - k$. Two functions with the same divisor differ by a constant, and distinct subsets give codewords with distinct supports, so we obtain every minimum-weight codeword exactly once up to scaling. Writing A_{n-k} for the number of codewords of weight $n - k$,

$$A_{n-k} = (q-1) \cdot \#\mathcal{S}, \quad \text{where}$$
$$\mathcal{S} = \Big\{ S \subseteq \operatorname{supp} D : \#S = k,\ \bigoplus\nolimits_{P \in S} P = \Sigma(G) \Big\}.$$

Counting the codewords of a given weight is hard in general. Here the group law reduces it to counting the solutions of a subset-sum problem in $\mathcal{E}(\mathbb{F}_q)$, which is the form in which the minimum distance of elliptic codes is usually studied [14].

A second consequence concerns the shape of G itself. Up to coordinate scalings, we may take G to be supported on up to two points, and every construction henceforth takes G in one of these two shapes.

Corollary 5 (Two-point normal form)

Fix D and G as in Section 3. Let $Q_0, Q_1 \in \mathcal{E}(\mathbb{F}_q) \setminus \operatorname{supp} D$ be such that $G' := (k-1)(Q_0) + (Q_1)$ satisfies $\Sigma(G') = \Sigma(G)$. Then $C_{\mathcal{L}}(D, G)$ and $C_{\mathcal{L}}(D, G')$ differ at most by coordinate scaling. In particular, they are equivalent codes.

Proof. See page 13 in the appendix. □

A third consequence is that the group law supplies the code with automorphisms. For $R \in \mathcal{E}(\mathbb{F}_q)$, the **translation** map $\tau_R : P \mapsto P \oplus R$ is an automorphism of $\mathcal{E}$ as a curve. It acts on functions by $\tau_R^* f = f \circ \tau_R$ and on divisors by $\tau_R^*(P) = (P \ominus R)$, so that $\tau_R^* \mathcal{L}(G) = \mathcal{L}(\tau_R^* G)$.

Theorem 6 (Translation automorphisms)

Let $C := C_{\mathcal{L}}(D, G)$ be as in Section 3. Suppose $R \in \mathcal{E}(\mathbb{F}_q)$ satisfies $[k]R = \infty$ and $\operatorname{supp} D$ is stable under τ_R.[a] Then τ_R induces a monomial automorphism of C.

[a] The latter condition holds, for instance, when $\operatorname{supp} D$ is a subgroup of $\mathcal{E}(\mathbb{F}_q)$ containing R, as in Theorem 7.

Proof. See page 13 in the appendix. □

These automorphisms are a feature of genus 1. Denote by $\operatorname{Aut}(\mathcal{E}, \infty)$ the subgroup of automorphisms fixing ∞, which has cardinality dividing 24 [4, Thm. III.10.1]. Then, the automorphism group of $\mathcal{E}$ as a curve is $\operatorname{Aut}\mathcal{E} = \mathcal{E}(\mathbb{F}_q) \rtimes \operatorname{Aut}(\mathcal{E}, \infty)$ [15], which has cardinality $\mathcal{O}(q)$. For a curve $\mathcal{C}$ of genus $g \geq 2$, $\#\operatorname{Aut}\mathcal{C}$ is bounded in terms of g alone, with $\#\operatorname{Aut}\mathcal{C} < 16g^4$ unless $\mathcal{C}$ is a Hermitian curve [16], so for fixed genus, it cannot keep pace with the number of rational points as q grows. As applications, Tong and Ding build quasi-cyclic near-MDS codes from elliptic curves using $\operatorname{Aut}(\mathcal{E}, \infty)$ [17], while Ma and Xing construct optimal locally repairable codes using $\operatorname{Aut}\mathcal{E}$ [15].

5 MDS elliptic codes and bounds

The question of whether a given AG code is MDS is ordinarily one about the vanishing of rational functions, settled by Riemann–Roch computations. For $g = 1$, it becomes a purely combinatorial question about a finite abelian group, since $C_{\mathcal{L}}(D, G)$ is MDS precisely when no k-element subset of $\operatorname{supp} D$ sums to $\Sigma(G)$. This section surveys constructions that arrange D and G so that this holds, and then asks how long such a code can be.

Elliptic curves were introduced in order to escape the length restriction on GRS codes, for which $n \leq q+1$. By the Hasse–Weil bound of Section 2, $\mathcal{E}(\mathbb{F}_q)$ may contain as many as $M := q + 1 + \lfloor 2\sqrt{q} \rfloor$ points. Being MDS, however, is expensive. As $\operatorname{supp} D$ grows, some k-subset of it is eventually forced to sum to $\Sigma(G)$, whereafter Theorem 4 precludes the MDS property. We shall see below that an MDS elliptic code cannot use more than roughly half of the available rational points. The full length $n \approx q + 2\sqrt{q}$ is available only to near-MDS codes.

The simplest way to obtain the condition of Theorem 4 is a parity argument on a one-point code. Suppose $N := \#\mathcal{E}(\mathbb{F}_q)$ is even and let $H \leq \mathcal{E}(\mathbb{F}_q)$ be a subgroup of index 2, so that $\mathcal{E}(\mathbb{F}_q) = H \sqcup (Q \oplus H)$ for any $Q \notin H$. Sums then behave predictably, as k elements of H sum into H, while k elements of $Q \oplus H$ sum into H or $Q \oplus H$ according to the parity of k.

Theorem 7 (Index-two codes [18])

Let $N = \#\mathcal{E}(\mathbb{F}_q)$ be even and let $H \leq \mathcal{E}(\mathbb{F}_q)$ be a subgroup of index 2. Choose $Q \in \mathcal{E}(\mathbb{F}_q) \setminus H$.

(a) If k is odd, set $D = \sum_{P \in H}(P)$ and $G = k(Q)$. Then $C_{\mathcal{L}}(D, G)$ is an $[N/2,\ k,\ N/2 - k + 1]_q$ MDS code.

(b) If k is even, choose $P \in H$ and set $D = \sum_{P' \in H \setminus \{P\}}(P')$ and $G = (k-1)(P) + (Q)$. Then $C_{\mathcal{L}}(D, G)$ is an $[N/2 - 1,\ k,\ N/2 - k]_q$ MDS code.

Proof. In both cases $\operatorname{supp} D \cap \operatorname{supp} G = \varnothing$ and $\deg G = k$, so the stated length and dimension are immediate.

For (a), the quotient $\mathcal{E}(\mathbb{F}_q)/H$ is generated by the coset $Q \oplus H$ and has order 2. As $\Sigma(G) = [k]Q$ with k odd, $\Sigma(G)$ lies in $Q \oplus H$.

For (b), $\Sigma(G) = [k-1]P \oplus Q$ with $[k-1]P \in H$, so again $\Sigma(G)$ lies in $Q \oplus H$.

In both cases, $\Sigma(G) \notin H$. But any k points of $\operatorname{supp} D = H$ sum into H, so Equation (2) fails and $C_{\mathcal{L}}(D,G)$ is MDS by Theorem 4. □

Extending by one coordinate lengthens the code in Theorem 7(b) to $[N/2,\ k,\ N/2-k+1]_q$ [19], in direct analogy with extended GRS codes. MDS codes of length $N/2$ therefore exist for every admissible k [18].

A second family of constructions produces MDS codes that are close to self-dual, which is what the quantum constructions of Section 6 require. A code C is **iso-dual** if $C^{\perp_E}$ can be obtained from C only by coordinate scalings, and **self-dual** if $C^{\perp_E} = C$. Both cases force $n = 2k$. The following construction of Zhu and Zhao produces iso-dual MDS elliptic codes over $\mathbb{F}_{2^s}$. Its MDS-ness again follows from Theorem 4, this time by an argument about element orders rather than parity.

Theorem 8 (Iso-dual MDS codes [20])

Let q, k, and $N = \#\mathcal{E}(\mathbb{F}_q)$ be even, so that $\mathcal{E}(\mathbb{F}_q)$ contains a point Q of order 2. Choose k distinct points $P_1,\dots,P_k \in \mathcal{E}(\mathbb{F}_q)^*$ of odd order where no two are inverse, and set

$$D = \sum_{i=1}^{k}\Big((Q \oplus P_i) + (Q \ominus P_i)\Big) \quad \text{and}$$
$$G = (k-1)(\infty) + (Q).$$

Then $C_{\mathcal{L}}(D,G)$ is an $[n,k,k+1]_q$ iso-dual MDS code, with $n = 2k \le \frac{N}{2} - 1$.

Proof. For the MDS property, let $\mathcal{E}_{\text{odd}} \le \mathcal{E}(\mathbb{F}_q)$ be the subgroup of points of odd order. The sum of any k points $Q \oplus P_{i_j}$ in $\operatorname{supp} D$ is

$$\bigoplus_{j=1}^{k}\Big(Q \oplus P_{i_j}\Big) = [k]Q \oplus \bigoplus_{j=1}^{k}\Big(\oplus P_{i_j}\Big) = \bigoplus_{j=1}^{k}\Big(\oplus P_{i_j}\Big) \in \mathcal{E}_{\text{odd}},$$

since k is even and $[2]Q = \infty$. Since $P_1,\dots,P_k \in \mathcal{E}_{\text{odd}}$, this sum has odd order and cannot equal $\Sigma(G) = Q \notin \mathcal{E}_{\text{odd}}$, so $C_{\mathcal{L}}(D,G)$ is MDS by Theorem 4.

For iso-duality, recall from Section 3 that $C_{\mathcal{L}}(D,G)^{\perp_E} = C_{\mathcal{L}}(D,H)$ with $H = D - G + \operatorname{div}\eta$. Recall from Section 2 that $\operatorname{div}\eta$ is principal. By Lemma 3, $\Sigma(\operatorname{div}\eta) = \infty$. Hence,

$$\deg H = \deg D - \deg G + 0 = 2k - k = k = \deg G.$$

Each pair of points $Q \oplus P_i$ in D sums to $[2]Q = \infty$, so $\Sigma(D) = \infty$ and

$$\Sigma(H) = \Sigma(D) \ominus \Sigma(G) \oplus \Sigma(\operatorname{div}\eta) = \ominus Q = Q = \Sigma(G),$$

the last step using that Q has order 2. By Corollary 5, $C_{\mathcal{L}}(D,H)$ and $C_{\mathcal{L}}(D,G)$ differ at most by coordinate scaling, which gives iso-duality of $C_{\mathcal{L}}(D,G)$.

For the bound on n, $Q \notin \mathcal{E}_{\text{odd}}$ implies $[\mathcal{E}(\mathbb{F}_q) : \mathcal{E}_{\text{odd}}] \ge 2$, so $\#\mathcal{E}_{\text{odd}} \le \frac{N}{2}$. Since $\operatorname{supp} D \subseteq Q \oplus (\mathcal{E}_{\text{odd}} \setminus \{\infty\})$, we have $n \le \#\mathcal{E}_{\text{odd}} - 1 \le \frac{N}{2} - 1$. □

Another important question to ask is how long an MDS elliptic code can be. Write $\mathrm{MEC}(k,q)$ for the maximum length of a nontrivial q-ary MDS elliptic code of dimension k, where we temporarily allow $\operatorname{supp} G$ to contain non-rational points. By Theorem 4, the problem is one in additive combinatorics. How large can a subset $\mathcal{T} \subseteq \mathcal{E}(\mathbb{F}_q)$ be before every point of the group is a sum of k distinct elements of $\mathcal{T}$? Once $\operatorname{supp} D$ is that large, no choice of G yields the MDS property.

The first bound is due to Munuera [21], who showed

$$\mathrm{MEC}(k,q) \le \frac{q+1}{2} + \sqrt{q} + k, \tag{3}$$

already confining an MDS elliptic code to about half of the rational points. This was improved by Li, Wan and Zhang [14] for large k, and by Han and Ren [22, 23] for small k, before the exact value was determined by Chen et al. [24], also for small k.

Theorem 9 (Maximal length [24, Table 1])

Let $q \ge 289$ and $3 \le k \le \frac{1}{10}\big(q + 1 - 2\sqrt{q}\big)$. Then

$$\mathrm{MEC}(k,q) = \left\lfloor \tfrac{M}{2} \right\rfloor.$$

If we further impose $2 \mid k$ and $\operatorname{supp} G \subseteq \mathcal{E}(\mathbb{F}_q)$, the maximal length is instead $\left\lfloor \frac{M}{2} \right\rfloor - 1$.

On a maximal curve with N even, so that $N = M$, both constructions of Theorem 7 meet this bound, part (b) in its form with $\operatorname{supp} G$ rational.

One may ask whether, given $\mathcal{E}$, we can bound the length of an MDS elliptic code on $\mathcal{E}$ using $\#\mathcal{E}(\mathbb{F}_q)$ instead of the generic Hasse–Weil upper bound M. This is clearly stronger than Theorem 9.

Conjecture 10 (Tighter MDS elliptic bound)

Let C be an MDS $[n,k]_q$ elliptic code on $\mathcal{E}$, with $N = \#\mathcal{E}(\mathbb{F}_q)$. If q (equivalently, N) is sufficiently large and $3 \le k \le n-3$, then $n \le \lfloor N/2 \rfloor$.[a]

[a] The bound on k excludes trivial cases.

Conjecture 10 is open in general, but proven for iso-dual codes with $N \ge 255$ [20, Thm. 3.1].

6 Quantum elliptic codes

Due to space constraints, we refer to [25–28] as references to quantum codes, EAQECCs, and undefined terminology in this section.

There are two major constructions of quantum codes from classical linear codes, using the Euclidean and Hermitian hulls respectively. This section explores applications of elliptic curves to both cases.

Write $\mathrm{Hull}_\bullet(C) := C \cap C^{\perp_\bullet}$ for the hull of C, where $\bullet \in \{\mathrm{E}, \mathrm{H}\}$ selects the Euclidean or Hermitian inner product. The extreme cases $\mathrm{Hull}_\bullet(C) = C$ and $\mathrm{Hull}_\bullet(C) = \{\mathbf{0}\}$ are the **self-orthogonal** and **linear complementary dual** (LCD) codes. An iso-dual code sits between the two, which makes Theorem 8 useful here.

For an EAQECC with parameters $[\![n, \kappa, d; c]\!]_q$, the Singleton bound reads

$$2d \le n - \kappa + c + 2 \quad \text{whenever} \quad d \le \tfrac{n}{2} + 1,$$

and an EAQECC achieving it is called **MDS** [29]. The Euclidean hull controls both κ and c.

Lemma 11 (Hull bound [30, Lem. 16])

Let C be an $[n,k,d]_q$ code and put $h = \dim \mathrm{Hull}_\mathrm{E}(C)$. Then, there exists an EAQECC with parameters $[\![n,\ k-h,\ d;\ n-k-h]\!]_q$.
If C is MDS and $d \le \frac{n}{2}+1$ (equivalently, $k \ge \frac{n}{2}$), this EAQECC is also MDS.

The construction in Theorem 8 yields MDS EAQECCs via Lemma 11. Table 13 lists parameters for some such EAQECCs over small fields $\mathbb{F}_{2^s}$. The dimension and the entanglement parameter agreeing is the signature of an EAQECC built from an iso-dual code.

Theorem 12 (MDS EAQECCs [20])

Let C be an $[n = 2k,\ k,\ k+1]_q$ iso-dual MDS code from Theorem 8. Put $h = \dim \mathrm{Hull}_\mathrm{E}(C)$ and $c = k - h$. Then $c \ge 1$, and there exists an MDS EAQECC with parameters $[\![n,\ c,\ k+1;\ c]\!]_q$. Moreover, $n \le \frac{M}{2} - 1$.

Proof. See page 13 in the appendix. □

q	Parameters	q	Parameters
16	$[\![8,4,5;4]\!]$	64	$[\![36,16,19;16]\!]$
32	$[\![20,8,11;8]\!]$	64	$[\![36,18,19;18]\!]$

Table 13. Parameters for some MDS elliptic EAQECCs over small fields of characteristic 2 [20, Ex. 6.1]

We note that nothing in Lemma 11 is special to genus 1. What elliptic curves contribute is the supply of iso-dual MDS codes in Theorem 8, and hence the MDS EAQECCs of Theorem 12.

We now turn our attention to Hermitian hulls, for which codes are taken over $\mathbb{F}_{q^2}$ with $\langle \mathbf{u}, \mathbf{v} \rangle_\mathrm{H} = \sum_{i=1}^n u_i v_i^q$. Classical Hermitian self-orthogonal (HSO) codes (i.e. those with $\mathrm{Hull}_\mathrm{H}(C) = C$) yield quantum stabiliser codes as follows.

Lemma 14 (Hermitian construction [31])

If C is an HSO $[n,k]_{q^2}$ code, then there exists a quantum code with parameters $[\![n,\ n-2k,\ \ge d(C^{\perp_\mathrm{H}})]\!]_q$.

The lemma is due to [32, 33]. Constructing HSO codes is a difficult problem in general, because unlike the Euclidean dual, the Hermitian dual has no closed form. Fortunately, one can circumvent this for AG codes using $C^{\perp_\mathrm{H}} = (C^{\perp_\mathrm{E}})^q$ [28, 31], where the exponent raises every coordinate to the qth power, which reduces the problem to the Euclidean case. For $\mathbf{v} \in (\mathbb{F}_{q^2}^*)^n$, write $\mathbf{v} \star C := \{(v_1 c_1, \ldots, v_n c_n) : \mathbf{c} \in C\}$.

Theorem 15 (HSO criterion [31, Thm. 3])

Let $C_\mathcal{L}(D,G)$ be an elliptic code over $\mathbb{F}_{q^2}$ as in Section 3. Let $\omega \in \Omega_\mathcal{E}$ satisfy $\mathrm{ord}_{P_i}(\omega) = -1$ for $1 \le i \le n$, and put $H = D - G + \mathrm{div}\,\omega$. If $qG \le H$ and $\mathbf{v} \in (\mathbb{F}_{q^2}^*)^n$ satisfies $v_i^{q+1} = \mathrm{Res}_{P_i}(\omega)$

for $1 \le i \le n$, then $\mathbf{v} \star C_{\mathcal{L}}(D,G)$ is HSO.

Proof. See page 13 in the appendix. □

Sok realises these conditions on a maximal curve in characteristic 2.

Construction 16 ([31, Constr. 5 & Thm. 6])

Let

- $q := 2^s$ for some $s \in \mathbb{Z}^+$,
- Tr be the absolute trace of $\mathbb{F}_{q^2}$ over $\mathbb{F}_2$,
- $c := 0$ if s is odd, else $c \in \mathbb{F}_{q^2}$ with $\mathrm{Tr}(c) = 1$,
- $\mathcal{E} : y^2 + y = x^3 + c$ over $\mathbb{F}_{q^2}$, a maximal curve,
- $n := \#\mathcal{E}(\mathbb{F}_{q^2})^* = q^2 + 2q$,
- $U := \{\alpha \in \mathbb{F}_{q^2} : \mathrm{Tr}(\alpha^3 + c) = 0\} = \{x(P) : P \in \mathcal{E}(\mathbb{F}_{q^2})^*\}$,
- $D := \sum_{i=1}^{n} (P_i) = \sum_{P \in \mathcal{E}(\mathbb{F}_{q^2})^*} (P)$,
- $G := k(\infty)$ for some $1 \le k \le q$,
- $h := \prod_{\alpha \in U} (x - \alpha)$, assuming $h'(U) \subseteq \mathbb{F}_q$, and
- $\mathbf{v} = (v_1, \ldots, v_n)$ with $v_i^{q+1} = 1/h'(x(P_i))$.

Then, $\mathbf{v} \star C_{\mathcal{L}}(D,G)$ is an HSO $[n,k]_{q^2}$ code by Theorem 15, yielding an $[\![n,\ n-2k,\ \ge k]\!]_q$ quantum code by Lemma 14.

We refer to [31, Constr. 5 & Thm. 6] for the proof that the conditions of Theorem 15 and Lemma 14 are met.[4]

The advantage of this construction is its length. A one-point code on an elliptic curve over $\mathbb{F}_{q^2}$ can use at most the $q^2 + 2q$ affine points allowed by Hasse–Weil, and Construction 16 uses all of them. This is possible because HSO imposes no subset-sum restriction on $\mathrm{supp}\, D$, so the barrier $\mathrm{MEC}(k,q)$ of Section 5 does not apply.

Construction 16 produces quantum codes with much better parameters than prior methods. The q-ary quantum codes previously obtained from elliptic curves [34] have length at most $q + \lfloor 2\sqrt{q} \rfloor - 5$ [31, Rem. 3], which is of the order of $\#\mathcal{E}(\mathbb{F}_q)$ rather than $\#\mathcal{E}(\mathbb{F}_{q^2})$. For $q = 4$, this caps the length at 3, against 24 using Construction 16. The price is at most one unit of distance. The quantum Singleton bound $2d \le n - \kappa + 2$ gives $d \le k + 1$, so $d \in \{k, k+1\}$, mirroring the near-MDS behaviour of classical elliptic codes in Section 3. Since the length $q^2 + 2q$ exceeds $q^2 + 1$, the conjectured maximal length of a q-ary quantum MDS code [31], we expect $d = k$ whenever $k \ge 2$.

7 Isogenies

Let $\mathcal{E}_i$ be an elliptic curve over $\mathbb{F}_q$, for $i = 1, 2$. An **isogeny** $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ is a nonconstant morphism of curves that fixes ∞. Such a map is automatically a surjective group homomorphism with finite kernel [4, §III.4]. An isogeny induces a **pullback** map φ^* on functions, differentials, and divisors, defined by

$$\begin{aligned} \mathbb{F}_q(\mathcal{E}_2) &\to \mathbb{F}_q(\mathcal{E}_1), & f &\mapsto f \circ \varphi \\ \Omega_{\mathcal{E}_2} &\to \Omega_{\mathcal{E}_1}, & \mathrm{d}f &\mapsto \mathrm{d}(f \circ \varphi) \\ \mathrm{Div}\,\mathcal{E}_2 &\to \mathrm{Div}\,\mathcal{E}_1, & \sum_{Q \in \mathcal{E}_2} n_Q (Q) &\mapsto \sum_{Q \in \mathcal{E}_2} n_Q \sum_{P \in \varphi^{-1}(Q)} e_P (P) \end{aligned}$$

where e_P denotes the ramification index of $P \in \mathcal{E}_1$ in $\varphi^{-1}(Q)$, the **fibre** over $Q \in \mathcal{E}_2$. Thus, φ induces a field extension $\varphi^* \mathbb{F}_q(\mathcal{E}_2) \le \mathbb{F}_q(\mathcal{E}_1)$, whose degree is denoted by $\ell := \deg \varphi$. An isogeny is called **separable** if this extension is separable, which we assume henceforth.[5] Additionally, there is a **pushforward** map φ_* on divisors and functions, defined by

$$\begin{aligned} \mathbb{F}_q(\mathcal{E}_1) &\to \mathbb{F}_q(\mathcal{E}_2), & f &\mapsto \left(\varphi_* f : Q \mapsto \prod_{P \in \varphi^{-1}(Q)} f(P) \right) \\ \mathrm{Div}\,\mathcal{E}_1 &\to \mathrm{Div}\,\mathcal{E}_2, & \sum_{P \in \mathcal{E}_1} n_P (P) &\mapsto \sum_{P \in \mathcal{E}_1} n_P \big(\varphi(P) \big), \end{aligned}$$

where $\varphi_* f$ is simply the algebraic norm and $(\varphi_* f)(Q)$ is defined whenever f is regular on $\varphi^{-1}(Q)$.

There are several ways that an isogeny φ can transform the code $C_{\mathcal{L}}(D,G)$ into another, namely

1. $C_{\mathcal{L}}(\varphi^* D, \varphi^* G) = \{ f(\varphi^* D) : f \in \mathcal{L}(\varphi^* G) \}$,
2. $C_{\mathcal{L}}(\varphi_* D, \varphi_* G) = \{ f(\varphi_* D) : f \in \mathcal{L}(\varphi_* G) \}$,
3. $\varphi^* C_{\mathcal{L}}(D,G) = \{ (\varphi^* f)(\varphi^* D) : f \in \mathcal{L}(G) \}$, or
4. $\varphi_* C_{\mathcal{L}}(D,G) = \{ (\varphi_* f)(\varphi_* D) : f \in \mathcal{L}(G) \}$,

where D and G live on $\mathcal{E}_2$ for the pullbacks and on $\mathcal{E}_1$ for the pushforwards.

[4] Note that our n and k are Sok's $2n$ and $k-1$ in [31], a consequence of our convention $n = \deg D$ and $k = \deg G$.

[5] Equivalently, $\deg \varphi = \#\ker \varphi$, or $e_P = 1 \ \forall P$.

This interaction is largely unexplored in the literature. We tasked Claude Opus 5.5 with writing SageMath code that constructs all four codes, and tested it on curves over small fields, notably a 3-isogeny from $\mathcal{E}_1 : y^2 = x^3 + 5$ to $\mathcal{E}_2 : y^2 = x^3 + 20$ over $\mathbb{F}_{31}$, whose 39 rational points form a cyclic group. The code can be found in [35]. Running the experiments, we observe the following.

- Method 4 gives nonlinear codes, as φ_* is multiplicative rather than additive on functions. The experiment with $k = 3$ yielded only 3110 distinct nonzero vectors, whereas a linear $[12, 3]_{31}$ code has $31^3 - 1 = 29790$.
- Method 2 gives invalid AG codes if $\operatorname{supp} D$ meets some fibre $\varphi^{-1}(Q)$ more than once, since $\varphi_* D$ then has a repeated point Q.
- Method 3 gives a subcode of method 1, as $\varphi^*\mathcal{L}(G) \subseteq \mathcal{L}(\varphi^* G)$. Since $(\varphi^* f)(P) = f(\varphi(P))$, it is the ℓ-fold repetition of $C_{\mathcal{L}}(D, G)$, with parameters $[\ell n, k, \ell d]$. For $k \geq 3$, it is not an elliptic code itself, as $\ell d \leq \ell(n - k + 1) < \ell n - k$ violates the bound of Section 3.
- Method 1 is the most interesting, and we study it for the rest of the section.

Denote by $\hat{\varphi} : \mathcal{E}_2 \to \mathcal{E}_1$ the dual isogeny, which satisfies $\hat{\varphi} \circ \varphi = [\ell]$ [4, §III.6]. A code is **quasi-cyclic** of index m if it is invariant under a coordinate permutation consisting of m disjoint cycles of equal length [36]. We observed the following properties of the code from method 1.

Theorem 17 (Pullback codes)

Let $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ be a separable isogeny with cyclic kernel $K := \ker \varphi \subseteq \mathcal{E}_1(\mathbb{F}_q)$. Let $C := C_{\mathcal{L}}(D, G)$ be an $[n, k]_q$ elliptic code on $\mathcal{E}_2$ with

$$D = \textstyle\sum_{i=1}^{n} (P_i) \in \operatorname{Div} \mathcal{E}_2, \quad \varphi^{-1}(P_i) \subseteq \mathcal{E}_1(\mathbb{F}_q).$$

Then $C' := C_{\mathcal{L}}(\varphi^* D, \varphi^* G)$ is an $[\ell n, \ell k, \geq \ell(n - k)]_q$ elliptic code on $\mathcal{E}_1$. Moreover,

(a) if $D - 2G$ is principal, then C and C' are both iso-dual,

(b) each $T \in K$ acts on C' by the coordinate permutation induced by $P \mapsto P \oplus T$, so C' is quasi-cyclic of index n (i.e. co-index ℓ),

(c) the subcode of C' fixed by K is $\varphi^* C$, the ℓ-fold repetition of C from method 3,

(d) suppose $\ell \mid q - 1$, and let $\zeta \in \mathbb{F}_q$ have order ℓ. Then, C' is monomially equivalent to the **matrix product code** $[C_0, \ldots, C_{\ell-1}] \cdot \mathbf{V}$ [19, 37] with $\mathbf{V} := [\zeta^{jm}]_{j,m=0}^{\ell-1}$, where each C_j is a scaling of the **twisted** code $C_{\mathcal{L}}(D, G_j)$.

Here, G_j is any divisor disjoint from D with $\deg G_j = k$ and $\Sigma(G_j) = \Sigma(G) \oplus [j]R$, where R is a suitable generator for $\langle R \rangle = \ker \hat{\varphi}$, and

(e) if C' is MDS, then no k points of $\operatorname{supp} D$ sum to an element of $\Sigma(G) \oplus \ker \hat{\varphi}$. In the setting of (d), every twist $C_{\mathcal{L}}(D, G_j)$ is then MDS.

See page 13 in the appendix for a proof of this result, which was supplied by Claude and verified by the authors, ChatGPT, and Gemini. To our knowledge, Theorem 17 does not appear in the literature in this form. However, its proof ingredients are standard in arithmetic geometry.

Theorem 17 has three immediate applications. The first is **longer iso-dual elliptic codes**. The hypothesis of (a) holds for the codes of Theorem 8, as $\Sigma(D) = \infty = [2]Q = \Sigma(2G)$. Method 1 therefore turns a short iso-dual MDS code of length $2k < \frac{N}{\ell}$ into a long iso-dual code of length $2\ell k$ with the same rate $\frac{1}{2}$ and relative designed distance $\frac{n-k}{n}$, but typically at the cost of the MDS property (i.e. 1 unit of distance).

The second is **decoding via twists**. Ling and Solé showed that every quasi-cyclic code decomposes into shorter constituent codes [36]. Part (d) identifies the constituents of C' as scalings of the twists $C_{\mathcal{L}}(D, G_j)$, which are $[n, k, \geq n - k]$ elliptic codes on $\mathcal{E}_2$. Decoding the ℓ twists separately, each up to half its minimum distance, corrects every error pattern meeting at most $\lfloor (n - k - 1)/2 \rfloor$ fibres $\varphi^{-1}(P_i)$, however many errors each fibre contains. Thus, errors concentrated on few fibres can be corrected by ℓ decoders of length n, rather than one of length ℓn.

The third is a supply of **quasi-cyclic elliptic codes**. Quasi-cyclic codes generalise cyclic codes and have a rich algebraic structure [36]. By (b), method 1 produces quasi-cyclic elliptic codes of co-index ℓ with

the same rate $\frac{k}{n}$ and relative designed distance $\frac{n-k}{n}$ as C, provided $\mathcal{E}_1(\mathbb{F}_q)$ has a cyclic subgroup of order ℓ. We can combine this with Theorem 7 to $\mathcal{E}_1$ to obtain **MDS quasi-cyclic elliptic codes**, provided that the index-two subgroup is compatible with the fibres of φ. This is illustrated in Construction 18 and example 19.

Construction 18 (MDS QC pullback codes)

Let $\varphi : \mathcal{E}_1 \to \mathcal{E}_2$ and K be as in Theorem 17, with ℓ odd. Let $N = \#\mathcal{E}_1(\mathbb{F}_q)$ be even, let $H \leq \mathcal{E}_1(\mathbb{F}_q)$ be a subgroup of index 2, and choose $Q \in \mathcal{E}_1(\mathbb{F}_q) \setminus H$.

(a) If k is odd, set $D = \sum_{P \in \varphi(Q \oplus H)}(P)$ and $G = k(\infty)$. Then $C' = C_{\mathcal{L}}(\varphi^* D, \varphi^* G)$ is an $[N/2,\ \ell k,\ N/2 - \ell k + 1]_q$ MDS code.

(b) For any k, set $D = \sum_{P \in \varphi(H) \setminus \{\infty\}}(P)$ and $G = (k-1)(\infty) + (\varphi(Q))$. Then C' is an $[N/2 - \ell,\ \ell k,\ N/2 - \ell - \ell k + 1]_q$ MDS code.

In both cases, C' is quasi-cyclic of co-index ℓ by Construction 18(b).

Proof. See page 14 in the appendix. □

Example 19 ($[140, 5k]_{251}$ MDS QC codes)

Consider $\mathcal{E}_1 : y^2 = x^3 + x + 100$ over $\mathbb{F}_{251}$. Then, $N = \#\mathcal{E}_1(\mathbb{F}_{251}) = 280$, which is 2 off from the upper Hasse–Weil bound $M = 282$. $(199, 0)$ is its only rational point of order 2, so $H := [2]\mathcal{E}_1(\mathbb{F}_{251})$ has index 2.

The point $T = (65, 14)$ has order $\ell = 5$, and Vélu's formulae give the isogeny $\varphi : \mathcal{E}_1 \to \mathcal{E}_2 : y^2 = x^3 + 231x + 13$ with kernel $\langle T \rangle$.

As $5 \mid 250 = q - 1$, we may take $\zeta = 20$, an element of order 5 in $\mathbb{F}_{251}$.

The 28 points of $\mathcal{E}_2$ whose fibres lie in $\mathcal{E}_1(\mathbb{F}_{251}) \setminus H$ are

(5,±17), (23,±26), (56,±8), (60,±31), (77,±52), (104,±24), (142,±124), (143,±67), (161,±100), (222,±7), (224,±85), (225,±5), (228,±75), and (232,±78).

Let D be their sum and $G = k(\infty)$ with k odd. By Construction 18, C' is a quasi-cyclic $[140, 5k, 141 - 5k]_{251}$ MDS code of co-index 5.

The kernel of $\hat{\varphi}$ is generated by $R = (141, 209)$, with multiples

$$[2]R=(20,121), \quad [3]R=(20,130), \quad \text{and} \quad [4]R=(141,42),$$

none of which lie in $\operatorname{supp} D$. The twisted divisors of Theorem 17(d) may be taken as

$$G_j = (k-1)(\infty) + ([j]R) \quad \text{for} \quad 0 \leq j \leq 4,$$

and C' is monomially equivalent to the matrix product code $[C_0, \ldots, C_4] \cdot \mathbf{V}$ with

$$\mathbf{V} = \left[\zeta^{jm}\right]_{j,m=0}^{4} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 20 & 149 & 219 & 113 \\ 1 & 149 & 113 & 20 & 219 \\ 1 & 219 & 20 & 113 & 149 \\ 1 & 113 & 219 & 149 & 20 \end{pmatrix},$$

where each C_j is a scaling of the twisted code $C_{\mathcal{L}}(D, G_j)$. By Theorem 17(e), every twist is a $[28, k, 29 - k]_{251}$ MDS code. For instance, for $k = 15$, C' is a $[140, 75, 66]_{251}$ code that glues together five $[28, 15, 14]_{251}$ MDS codes.

Taking instead D to be the sum of the 55 affine points of $\varphi(\mathcal{E}_1(\mathbb{F}_{251}))$, Theorem 17 gives quasi-cyclic $[275, 5k]_{251}$ codes for $1 \leq k \leq 54$. By Equation (3), the codes are near-MDS, with minimum distance $275 - 5k$.

8 Conclusion

This report has demonstrated that amongst all AG codes, the genus 1 case possesses unique properties that make it stand out from other genera. The group law of an elliptic curve supplies many beautiful arguments that are used to achieve better code parameters and desirable code properties. We summarise these below.

1. Showing whether a code is MDS reduces to the subset-sum criterion (Theorem 4), an algebraic combinatorial problem. This was used to construct iso-dual MDS codes (Theorem 8), which yielded MDS EAQECCs (Theorem 12).
2. Up to equivalence, we need only consider one- or two-point codes (Corollary 5).
3. Code automorphisms are abundant, arising from translations (Theorem 6) as well as from $\operatorname{Aut}(\mathcal{E}, \infty)$.
4. One has a strategy for constructing Hermitian self-orthogonal elliptic codes (Construction 16), which has resulted in improved parameters for

quantum codes.

5. Pulling back along an isogeny produces longer elliptic codes that preserve the rate and relative designed distance, inherit iso-duality, are quasi-cyclic, and split into twisted elliptic codes (Theorem 17).

However, the current research landscape focusing specifically on elliptic codes is relatively limited, compared to other families of AG codes. Open questions include

- the development of an efficient and effective decoding algorithm,
- the resolution of Conjecture 10, and
- further constructions of Hermitian self-orthogonal or dual-containing codes.

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Appendix

Proof of Theorem 4. Since $\deg G = k$ with $1 \le k \le n-1$, we have $\ell(G) = k > 0$ and $d^* = n - k > 0$, so [8, Rem. 2.2.5] applies, giving $d = n - k$ if and only if there is a divisor $D' \in \operatorname{Div}\mathcal{E}$ with $0 < D' \le D$, $\deg D' = k$ and $\ell(G - D') > 0$. As D is a sum of distinct points, such a D' is exactly a sum $(P_{i_1}) + \cdots + (P_{i_k})$ of k distinct points of $\operatorname{supp} D$.

For such a D', we have $\deg(G - D') = 0$, so $\ell(G - D') > 0$ if and only if $G - D'$ is principal [8, Cor. 1.4.12]. By Lemma 3, this occurs if and only if $\Sigma(G - D') = \infty$, if and only if $P_{i_1} \oplus \cdots \oplus P_{i_k} = \Sigma(G)$. □

Proof of Corollary 5. Both G and G' have degree k and sum $\Sigma(G)$, so $G'-G$ has degree 0 and sum ∞. By Lemma 3, $G'-G$ is principal, say $G' = G + \operatorname{div} u$ with $u \in \overline{\mathbb{F}}_q(\mathcal{E})^*$. Since $\operatorname{div} u$ is invariant under Frobenius σ, the functions $\sigma(u)$ and u have the same divisor, so $\sigma(u) = cu$ for some constant $c \in \overline{\mathbb{F}}_q^*$. Choose $\lambda \in \overline{\mathbb{F}}_q^*$ with $\lambda^{q-1} = c^{-1}$. Then $\sigma(\lambda u) = \lambda^q c u = \lambda u$, so replacing u by λu we may assume $u \in \mathbb{F}_q(\mathcal{E})^*$.

Both G and G' are disjoint from $\operatorname{supp} D$, so $\operatorname{div} u = G' - G$ has coefficient 0 at each P_i. Hence $\operatorname{ord}_{P_i}(u) = 0$ and $u(P_i) \in \mathbb{F}_q^*$. For $f \in \mathbb{F}_q(\mathcal{E})^*$ we have $\operatorname{div}(uf) = \operatorname{div} f + G' - G$, so $f \in \mathcal{L}(G')$ if and only if $uf \in \mathcal{L}(G)$, that is, $\mathcal{L}(G') = u^{-1}\mathcal{L}(G)$. Evaluating at D, every $f \in \mathcal{L}(G')$ satisfies $f(D) = \mathbf{v} \star (uf)(D)$ with $\mathbf{v} := \left(u(P_i)^{-1}\right)_{i=1}^n \in (\mathbb{F}_q^*)^n$, and uf ranges over all of $\mathcal{L}(G)$. Therefore

$$C_{\mathcal{L}}(D, G') = \mathbf{v} \star C_{\mathcal{L}}(D, G),$$

so the two codes differ only by coordinate scaling, and in particular are equivalent. This is the content of [8, Prop. 2.2.14]. □

Proof of Theorem 6. Both $\tau_R^* G$ and G have degree k, and

$$\Sigma(\tau_R^* G) = \Sigma(G) \ominus [k]R = \Sigma(G),$$

so $\tau_R^* G = G + \operatorname{div} u$ for some $u \in \mathbb{F}_q(\mathcal{E})^*$ by Lemma 3. Then $f \mapsto u \cdot \tau_R^* f$ is a linear bijection from $\mathcal{L}(G)$ to itself. Write $P_{\sigma(i)} := P_i \oplus R$, which lies in $\operatorname{supp} D$ by stability. Evaluating at D, the map sends $f(D)$ to $\left(u(P_i) f(P_{\sigma(i)})\right)_{i=1}^n$, which is a coordinate permutation of $f(D)$ followed by a scaling, and it maps C onto itself. □

Proof of Theorem 12. C satisfies the conditions of Lemma 11, yielding an MDS EAQECC with parameters $[\![n, k-h, d; n-k-h]\!]_q$. The dimension and the entanglement parameter both equal c. The bound on n is inherited from Theorem 8. Lastly, Zhu and Zhao bound $h \le k-1$ [20, Cor. 4.1], which gives $c \ge 1$. □

Proof of Theorem 15. Since $\operatorname{ord}_{P_i}(\omega) = -1$, we have $C_{\mathcal{L}}(D, G)^{\perp_{\mathrm{E}}} = \mathbf{r} \star C_{\mathcal{L}}(D, H)$ with $r_i = \operatorname{Res}_{P_i}(\omega)$ [8, Prop. 2.2.10].[6] Let $f, g \in \mathcal{L}(G)$. Then $g^q \in \mathcal{L}(qG) \subseteq \mathcal{L}(H)$, so

$$\begin{aligned}\left\langle \mathbf{v} \star f(D), \mathbf{v} \star g(D) \right\rangle_{\mathrm{H}} &= \sum_{i=1}^n r_i f(P_i) g^q(P_i) \\ &= f(D) \cdot \left(\mathbf{r} \star g^q(D)\right) = 0. \quad \square\end{aligned}$$

Proof of Theorem 17. We use three facts. First, $\varphi(P \oplus T) = \varphi(P)$ for $T \in K$, so $\tau_T^* \varphi^* = \varphi^*$. Second, $\mathbb{F}_q(\mathcal{E}_1)/\varphi^*\mathbb{F}_q(\mathcal{E}_2)$ is Galois with group $\{\tau_T^* : T \in K\}$ [4, Thm. III.4.10], and these automorphisms are defined over $\mathbb{F}_q$ as K is rational. Hence the functions fixed by every τ_T^* are exactly the pullbacks. Third, $\varphi^*(Q)$ is the sum of the ℓ distinct points of the fibre over Q, and distinct points have disjoint fibres. Comparing coefficients, $\varphi^* A \ge \varphi^* B$ if and only if $A \ge B$, and a divisor on $\mathcal{E}_1$ whose coefficients are constant on cosets of K is a pullback.

Parameters. The fibres over the points of D are n disjoint cosets of K in $\mathcal{E}_1(\mathbb{F}_q)$, so $\varphi^* D$ is a sum of ℓn distinct rational points, disjoint from $\operatorname{supp} \varphi^* G$. As $\deg \varphi^* G = \ell k$, the code C' is elliptic with the stated parameters by Section 3.

(a) Since $\varphi^* \operatorname{div} f = \operatorname{div} \varphi^* f$, if $D - 2G$ is principal, then so is $\varphi^* D - 2\varphi^* G$. It therefore suffices to show that an elliptic code $C = C_{\mathcal{L}}(D, G)$ with $D - 2G$ principal is iso-dual, as this applies to C' too. By Section 3, $C^{\perp_{\mathrm{E}}} = C_{\mathcal{L}}(D, H)$ with $H = D - G + \operatorname{div} \eta$, where $\operatorname{div} \eta$ is principal. Then $H \sim D - G \sim G$, and $C_{\mathcal{L}}(D, H)$ is a scaling of C [8, Prop. 2.2.14].

(b) By the first fact, τ_T^* fixes $\varphi^* D$ and $\varphi^* G$, so $f \mapsto \tau_T^* f$ maps $\mathcal{L}(\varphi^* G)$ onto itself. As $(\tau_T^* f)(P) = f(P \oplus T)$, this permutes the coordinates of each codeword within each fibre. If $K = \langle T \rangle$, the permutation consists of n disjoint ℓ-cycles, one per fibre.

(c) By the third fact, $\varphi^* \mathcal{L}(G) \subseteq \mathcal{L}(\varphi^* G)$, and each $\varphi^* f$ is constant on fibres. So, $\varphi^* C \subseteq C'$ is fixed by K and is the ℓ-fold repetition of C. Conversely, suppose $f(\varphi^* D)$ is fixed by K. For $T \in K$, the function $\tau_T^* f - f \in \mathcal{L}(\varphi^* G)$ vanishes at the $\ell n > \deg \varphi^* G$ points of $\varphi^* D$, so $\tau_T^* f = f$. By the second fact, $f = \varphi^* g$, and $g \in \mathcal{L}(G)$ by the third fact.

(d) Let $K = \langle T \rangle$, write $\tau := \tau_T^*$, and fix $\zeta \in \mathbb{F}_q$ of order ℓ. As $\tau^\ell = 1$ and $\ell \nmid q$, the operator τ on $\mathcal{L}(\varphi^* G)$ is diagonalisable with eigenvalues among the ζ^j. Let L_j

[6] This generalises the case $\omega = \eta$ of Section 3.

be the ζ^j-eigenspace and W_j its image in C', so that $C' = \bigoplus_j W_j$, since evaluation at $\varphi^* D$ is injective.

By Dedekind's lemma, $1, \tau, \dots, \tau^{\ell-1}$ are linearly independent over $\mathbb{F}_q(\mathcal{E}_1)$, and so over $\varphi^*\mathbb{F}_q(\mathcal{E}_2)$. Thus, τ has minimal polynomial $X^\ell - 1$ on $\mathbb{F}_q(\mathcal{E}_1)$. Hence, there is $u \in \mathbb{F}_q(\mathcal{E}_1)^*$ with $\tau(u) = \zeta u$. Then, $\operatorname{div} u$ is fixed by K, so $\operatorname{div} u = \varphi^* A$ for some $A \in \operatorname{Div}\mathcal{E}_2$ by the third fact. Also, u^ℓ is fixed by τ, so $u^\ell = \varphi^* h$ for some h, and $\varphi^*(\ell A) = \varphi^* \operatorname{div} h$ gives $\ell A = \operatorname{div} h$. In particular, $\deg A = 0$. By the approximation theorem [8, Thm. 1.3.1], replacing u by $u \cdot \varphi^* a$ for suitable $a \in \mathbb{F}_q(\mathcal{E}_2)^*$, we may assume $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$.

Put $R := \Sigma(A) \in \mathcal{E}_2(\mathbb{F}_q)$. As $\varphi^* A = \operatorname{div} u$ is principal and $\deg A = 0$, we have $\Sigma(\varphi^* A) = \hat{\varphi}(\Sigma(A))$ by the construction of $\hat{\varphi}$ [4, §III.6], so $R \in \ker \hat{\varphi}$. If $[m]R = \infty$ with $m < \ell$, then $mA = \operatorname{div} a$ for some a. Consider the function $v := u^m/\varphi^* a$. Its divisor is $\operatorname{div} v = m\varphi^*(A) - \varphi^*(mA) = 0$, so v is a nonzero constant, hence fixed by τ. But $\tau(v) = (\zeta u)^m/\varphi^* a = \zeta^m v$, which forces $\zeta^m = 1$, a contradiction. Thus, R has order ℓ. As $\hat{\varphi} \circ \varphi = [\ell]$ is separable for $\ell \nmid q$, so is $\hat{\varphi}$, and $\#\ker\hat{\varphi} = \ell$. Hence R generates $\ker\hat{\varphi}$, which is therefore rational.

Now, $f \in L_j$ if and only if f/u^j is fixed by τ, that is, $f = u^j \varphi^* g$ for some $g \in \mathbb{F}_q(\mathcal{E}_2)$. By the third fact, $\operatorname{div} f = \varphi^*(jA + \operatorname{div} g) \geq -\varphi^* G$ if and only if $g \in \mathcal{L}(G + jA)$. So $L_j = u^j \varphi^* \mathcal{L}(G + jA)$. Fix $\tilde{P}_i \in \varphi^{-1}(P_i)$ for each i, so that the fibre over P_i is $\{\tilde{P}_i \oplus [m]T : 0 \leq m < \ell\}$, and order the coordinates of C' as $\tilde{P}_1, \dots, \tilde{P}_n$, $\tilde{P}_1 \oplus T, \dots, \tilde{P}_n \oplus T$, …. For $f = u^j \varphi^* g \in L_j$, we have $f(\tilde{P}_i) = u(\tilde{P}_i)^j g(P_i)$, where u has neither a zero nor a pole at $\tilde{P}_i$ as $\operatorname{supp} A \cap \operatorname{supp} D = \varnothing$. Hence, the restrictions of W_j to the coordinates $\tilde{P}_1, \dots, \tilde{P}_n$ form $C_j := \mathbf{v}_j \star C_{\mathcal{L}}(D, G + jA)$, where $\mathbf{v}_j = (u(\tilde{P}_i)^j)_{i=1}^n$. Moreover, $f(\tilde{P}_i \oplus [m]T) = (\tau^m f)(\tilde{P}_i) = \zeta^{jm} f(\tilde{P}_i)$, so

$$W_j = \{(\mathbf{c} \mid \zeta^j \mathbf{c} \mid \cdots \mid \zeta^{j(\ell-1)} \mathbf{c}) : \mathbf{c} \in C_j\}.$$

Summing over j gives $C' = \bigoplus_j W_j = [C_0, \dots, C_{\ell-1}] \cdot V$. Finally, $G + jA$ is disjoint from D, with degree k and sum $\Sigma(G) \oplus [j]R$. By Lemma 3, any G_j as in the statement is equivalent to it, so $C_{\mathcal{L}}(D, G_j)$ is a scaling of $C_{\mathcal{L}}(D, G + jA)$ [8, Prop. 2.2.14].

(e) By the construction of $\hat{\varphi}$ [4, §III.6], every divisor $B \in \operatorname{Div}\mathcal{E}_2$ of degree 0 satisfies $\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B))$. Put $s_K := \Sigma(\varphi^*(\infty)) = \bigoplus_{T \in K} T$. Applying this to $B - b(\infty)$, every divisor B of degree b satisfies

$$\Sigma(\varphi^* B) = \hat{\varphi}(\Sigma(B)) \oplus [b] s_K.$$

Suppose that distinct points $P_{i_1}, \dots, P_{i_k} \in \operatorname{supp} D$ satisfy $P_{i_1} \oplus \cdots \oplus P_{i_k} \ominus \Sigma(G) \in \ker\hat{\varphi}$. Applying the identity to $B = \sum_j (P_{i_j})$ and to G, both of degree k, the ℓk points of the fibres over $P_{i_1}, \dots, P_{i_k}$ sum to

$$\hat{\varphi}\left(P_{i_1} \oplus \cdots \oplus P_{i_k}\right) \oplus [k] s_K = \hat{\varphi}(\Sigma(G)) \oplus [k] s_K = \Sigma(\varphi^* G).$$

These are ℓk distinct points of $\operatorname{supp} \varphi^* D$, so C' is not MDS by Theorem 4. In the setting of (d), we have $\ker\hat{\varphi} = \langle R \rangle$, so $\Sigma(G_j) = \Sigma(G) \oplus [j]R$ runs over $\Sigma(G) \oplus \ker\hat{\varphi}$. Hence, if C' is MDS, no k points of $\operatorname{supp} D$ sum to any $\Sigma(G_j)$, and each twist $C_{\mathcal{L}}(D, G_j)$ is MDS by Theorem 4. □

Proof of Construction 18. The image of K in $\mathcal{E}_1(\mathbb{F}_q)/H \cong \mathbb{Z}/2\mathbb{Z}$ has order dividing both $\#K = \ell$ and 2, so it is trivial, and $K \subseteq H$. Hence, H and $Q \oplus H$ are unions of cosets of K, i.e. they are partitioned by the fibres of φ. So $\varphi^{-1}(\varphi(H)) = H$ and $\varphi^{-1}(\varphi(Q \oplus H)) = Q \oplus H$. In particular, $\varphi(H) \cap \varphi(Q \oplus H) = \varnothing$. Also, every fibre over a point of $\operatorname{supp} D$ is rational, so Theorem 17 applies. As $\infty = \varphi(\infty) \in \varphi(H)$ and $\varphi(Q) \in \varphi(Q \oplus H)$, the supports of D and G are disjoint in both cases. Writing $K = \langle T \rangle$, we have $s_K := \sum_{T' \in K} T' = [\ell(\ell-1)/2]T = \infty$, since ℓ is odd.

For (a), $\Sigma(\varphi^* G) = [k] s_K = \infty \in H$. But $\operatorname{supp} \varphi^* D = Q \oplus H$, so any ℓk of its points sum into $[\ell k]Q \oplus H = Q \oplus H$, as ℓk is odd.

For (b), the fibre over $\varphi(Q)$ is $Q \oplus K$, so $\Sigma(\varphi^* G) = [k-1] s_K \oplus [\ell] Q \oplus s_K = [\ell] Q$, which lies in $Q \oplus H$ as ℓ is odd. But $\operatorname{supp} \varphi^* D \subseteq H$, so any ℓk of its points sum into H.

In both cases, no ℓk points of $\operatorname{supp} \varphi^* D$ sum to $\Sigma(\varphi^* G)$, so C' is MDS by Theorem 4. The lengths are $\ell \cdot \#\operatorname{supp} D$, namely $\ell \cdot \frac{N}{2\ell}$ and $\ell \cdot \left(\frac{N}{2\ell} - 1\right)$. Removing points from D preserves both arguments.

Note that the argument extends to a subgroup H of any prime index $m \nmid \ell$, giving MDS codes of length N/m whenever $m \nmid \ell k$. This also covers even ℓ, for example $\ell = 4$ with $m = 3$. □

AI prompts

At the beginning, we used Claude to evaluate our potential project directions. We then drafted a section-by-section outline of the report, which Claude referenced to search for relevant and recent publications, suggest results to include, evaluate their suitability for the report, and collate a bibliography containing those results.

- "Sort these project directions on elliptic codes into foundational, doable at undergraduate level, stretch, and novel, with research questions for each."
- "Validate or disprove our comments with research. We want to survey recent constructions such as iso-dual elliptic codes, study conditions for an elliptic code to be MDS, and include a short exploration of isogenies."
- "Suggest a few constructions that exploit the group law and are easiest to explain, avoiding references that need too much machinery. Motivate them by the fact that only elliptic curves have a group law."

Sections 1 to 3 were fully human-written (other than grammatical fixes). For Sections 4 to 6, we first reviewed the original papers that introduced the constructions that Claude suggested, and selected a few relevant ones. We then wrote a skeleton of Sections 4 to 6 based on those constructions and used Claude to expand the draft by filling in details such as theorems and references. Of course, we reviewed and revised the drafts.

- "Proofread Sections 1 to 3, and expand the draft of Sections 4 and 5 in the same tone and style."
- "Draft Section 6, focusing on applications rather than intuition. Expand the Hermitian self-orthogonal part after reading Lemma 6, Construction 5 and Theorem 6 of [31]."
- "Add one or two paragraphs to Section 4 on code automorphisms arising from the group law."

Ideation for Section 7 was done without AI assistance. After hypothesising the outcome of the four methods, we compared our hypotheses against Claude's own predictions. Claude then generated SageMath code to construct our codes and test our hypotheses. After running the code, we combined our observations into Theorem 17 and asked Claude to prove it in a more general case. Claude also wrote code to compute Example 19.

- "Predict the outcome of each of the four isogeny transformations, and suggest how to analyse them in SageMath, with a sample code file to test each one."
- "Suggest applications of method 1."
- "Prove the properties of method 1: its parameters $[\ell n, \ell k, \geq \ell(n-k)]$, inherited iso-duality, quasi-cyclicity, and its splitting into ℓ twisted elliptic codes."

We fed the statement and proof of Theorem 17 into ChatGPT and Gemini to verify whether the proof was correct. While all agreed that the proof was valid, we refined the proof based on their feedback to make certain arguments clearer.

We used Claude to proofread our manuscript for errors and missing hypotheses in results.

- "Why does Conjecture 10 require $3 \leq k \leq n-3$, and why does the bound in Theorem 12 differ from $\mathrm{MEC}(k,q)$?"
- "Proofread the changes to the document, including Section 7."
- "Confirm that the rewritten statement of Theorem 17 omits no crucial hypothesis, and whether its proof is original."

Lastly, we used Claude Code to expedite the process of writing and debugging the Manim code for our presentation.

- "Next slide. Unwrite all text, leaving only the curve. Then, write a Tex "AG Code", bolded, in the top left. Add 3D axes centred at the curve. Rotate the camera to give a 3D view of the graph."
- "Next slide. Animate the points on E1 moving to their images on E2 under the chosen isogeny φ. Add green lines between the image points as before to show that the group law is preserved."